

Interview questions for Database Administrator

1. During a database migration how would you handle data loss?

Data loss is one of the situations where database administrators experience the most pressure, especially if a migration project falls behind schedule. This question will help you learn more about how a candidate behaves under pressure and stress, and more importantly, what strategies they use to overcome this situation — understanding how to troubleshoot and recover data is a crucial function all DBA roles should know how to handle.

2. We need to build a new database for our employee records. How would you define the system storage requirements?

Ideally, the candidate will start answering this question by understanding what your current database situation looks like. Once having a grasp of this, they can start proposing and defining the system storage requirements. There's not a right or wrong answer. It depends on your company, but more importantly, what you're looking for with this question is to learn how a candidate, if this were a real situation, will respond to your demands and what database solutions they propose based on their knowledge and experience.

3. Tell me about your process for troubleshooting database problems

Troubleshooting is a regular task for a most DBA, so it's crucial to hire one that is able to do it efficiently. They need to answer by telling you what their process for identifying and addressing the issue is like. You should look for a solid process; if a candidate gives a general answer, this might be a red flag. Try to understand in detail how this process is like and how they use their available resources/tools, and their experience.

4.What measurements would you take to protect our databases from external threats?

Cyber attacks have existed since the beginning of the internet era. Now that the digital environment keeps increasing, protecting your company's and client's data becomes even more relevant. This question attains that urgency and a candidate should be able to give you different strategies to protect your databases from any type of threats. Whether it's by setting an HTTPS server, using real-time database monitoring, or deploying data encryption protocols, the answer can vary depending on a candidate's knowledge and experience.

5. What types of databases do you work with?

There are many different types of database infrastructures, so with this question, you'll be confirming if a candidate has the background you're looking for. For example, if your database is built with MySQL or Oracle, then it's important to see how familiar candidates are with these database servers and their level of expertise.

6.What is SQL Agent?

SQL Agent is the job scheduling mechanism in SQL Server. Jobs can be scheduled to run at a determined time or when a specific event happens. Also, jobs can be executed on demand. SQL Agent is popularly used to schedule administrative jobs such as backups.

7. What is DBCC?

DBBC statements are Database Console Commands and have four different areas:

1. Maintenance commands: Are those commands that allow the database administrator to perform maintenance activities (for example shrinking a file.)
2. Informational commands: Provide feedback regarding the database.
3. Validation commands: Include commands that validate the database such as the ever-popular CHECKDB.
4. Miscellaneous commands: Are those who can't be categorized in the three previous areas. They include statements like DBCC help.

8. Explain what a system database and a user database are

A system database is the default database installed when the SQL server is installed. There are 4 system databases: Master, MSDB, TempDB, and Model. A user database, on the other hand, is a database created to store data and start working with the data.

9. What are the operating modes in which Database Mirroring runs? What are the differences between them?

Database Mirroring runs in 2 operating modes: High-Safety Mode and High-Performance Mode. The first one (High-Safety Mode) ensures that the Principal and Mirrored database are in a synchronized state; that is the transactions are committed at the same time on both servers to ensure consistency. The second one (High-Performance Mode) ensures that the Principal database runs faster by not waiting for the Mirrored database to commit the transactions.

10. Explain the purpose of a model database

The model database consists of the template for all databases created in the SQL system. If the initial model database is modified, all subsequent databases created in the systems will reflect the changes. However, databases created earlier will not.

11. What is the difference between a data processor and a data controller?

The legislation applies to two different types of data-handlers: processors and controllers. According to the definitions provided by the General Data Protection Regulation (GDPR), a controller is the entity that determines the purpose, conditions, and means of the processing of personal data. On the other hand, the processor is an entity that processes personal data on behalf of the controller.

12. Explain what are the Right of Access requests

The Right of Access request (Art.15 – GDPR) provides individuals with the ability to know if a data controller is processing their personal data and, if that's the case, what that information is and why it is being processed.

Individuals are entitled to a copy of the personal data in question, which doesn't necessarily depend if someone is an employer, worker, or self-employed.

The DDPR contains exceptions that mostly focus on questions of public interest (like the investigation of crime).

Moreover, the GDPR recital 47 states that fraud prevention is a legitimate interest to process personal data: “The processing of personal data is strictly necessary for the purpose of preventing fraud also constitutes a legitimate interest of the data controller concerned”. Even for the purpose of fraud, the controller still has to prove that legitimate interest applies and that the processing of personal data is necessary.

13. Do all businesses need a Data Protection Officer (DPO)?

A DPO must be appointed in three specific cases:

- 1) public authorities,
- 2) organizations that engage in large-scale systematic monitoring, or
- 3) organizations that engage in large-scale processing of sensitive personal data.

If your organization doesn't fall into these categories, it's not necessary to appoint a DPO.

14. Can anyone access the personal data within your company? Or are there different levels of access?

As a controller or processor, you're entitled to process the data. However, this doesn't mean that all employees can access it – data should be available only for those in the company whose position requires them to have those rights. There are different levels of access; while some people might have full access with the rights of modifying or erasing data, others will only be able to view data.

15. How can you collect data (by email, activity tracking, etc.)?

It's a necessary step to check the different ways you can collect data. However, you firstly need to audit your data flows properly and verify you're satisfying all the legal obligations, such as obtaining proper consents for processing, providing the data subject with the information specified by GDPR, etc.

16. How did you learn about new applications/resources?

Data administrators, and any other IT roles, have to be on top of the new resources that appear on the market. The industry is constantly changing, and a big mistake would be to hire a DBA that is still working on old database versions. Make sure they know about all the changes whether because they are part of communities, Facebook groups, or simply because they keep informed all the time.

17. What is the most challenging project you've worked on? Why was it challenging and what was your role?

A project can be challenging for different reasons. Maybe they had a hard time coping with some team members, or maybe, because of an external situation they struggled with deadlines. Although this is a common question, it's a great way of learning more about a candidate and their soft skills.

18. Can you tell me of a time you made a mistake and how did you resolve it? Was there any way you could prevent it?

Everyone is human. If you face a candidate giving you the perfect, straight-out-of-the-book answers, believe it or not, that might be a red flag. Instead of the

“No, I haven’t made any mistakes, I’m perfect” kind of answer, look for someone honest. The trick of this question isn’t only about the mistake itself but also about the ‘how.’ Learning how they worked under pressure and how they solved it. Additionally, knowing how to prevent it it’s also a good sign that they understood exactly what had happened, and because of this, it’s very unlikely it will happen to them again.

19. If you had to work in a team with people with difficult personalities, and constant conflict, how would you manage it?

You can have a strong personality with clear leadership skills and critical thinking but still, be able to be empathic and listen to the other team members. Whether your candidate has a strong or a more relaxed personality, what you’re looking to learn with this question is how they cope with different opinions. It will help you better understand a candidate’s personality traits and how they might deal with conflict.

20. Imagine that you have to design a database system for an important client. You have a tight deadline, so all the team is working hard. However, you’ve noticed that as your boss is more worried about the deadline, he has ignored certain aspects. For example, not using SQL facilities to protect data integrity and he has even decided not to conduct regular testings to speed up the process. How would you handle this?

Ideally, your team doesn’t have this type of leader. However, this imaginary leader can be, in fact, a colleague who doesn’t care about the quality of a project but cares more about finishing it to get a paycheck. This question can help you understand the priorities a candidate has and how they behave around this

environment. Whether they speak up for what's right or simply agree that the boss is the master and commander, and if he says he's right, he is right.

21. What is your experience working with database servers?

What you're looking for in this question is to learn more about a candidate's familiarity with database servers, how they have used each system, whether it was Oracle or Microsoft or any other, and the years of experience they've had working with them.

22. What is the highest number of database servers you have worked with?

If you're a large organization, this question is perfect for you. One thing is working for a small business that has a small data center. A whole different thing is having to deal with a data center of a large corporation. If the applicant has only worked with small data centers, then you're facing the risk of them not knowing how to handle yours or facing constant problems. It's important to learn more about the sizes of their previous companies, the number of servers, and their environments.

23. Have you worked with on-premises databases, cloud databases or both?

Most businesses are currently moving to cloud infrastructure. If that's your case, then it's important to learn more about a candidate's experience working with cloud databases and know what environment they work best. Additionally, suppose your organization has an on-premises database and is looking to shift towards a hybrid database. In that case, you need to find out if a candidate will be able to support your long-term goals.

24. Why did you choose database administration?

Passion is what makes people thrive. But not everyone has an inspiring, amazing story about how they chose a specific career. It's interesting, however, to learn more about a candidate's background and, in this case, learn how they bumped into this career and whether they really enjoy it or not. You can also ask about their education and where they learned about database management.

25. Describe your workflow without direct supervision

Unless you're a micromanager on top of everyone at the office, looking for someone who can work independently is important. Moreover, when hiring developers who work remotely, they need to work without direct supervision all the time. Learn about their workflow and their working process, if they are organized and have a clear agenda of their priorities, or if they expect their leaders every day to tell them what to do.

26. What are the different types of normalization?

- The different types of normalization include:
 - First Normal Form (1NF): Eliminates repeating groups by ensuring each field contains only atomic values.
 - Second Normal Form (2NF): Achieves 1NF and ensures all non-key attributes are fully functionally dependent on the primary key.
 - Third Normal Form (3NF): Achieves 2NF and removes transitive dependencies.

- Boyce-Codd Normal Form (BCNF): A stricter version of 3NF where every determinant is a candidate key.

- Fourth Normal Form (4NF): Eliminates multi-valued dependencies.

- Fifth Normal Form (5NF): Removes any remaining anomalies.

27. How do you optimize a SQL query?

- Query optimization can be done by:

- Using indexes effectively.

- Avoiding SELECT and specifying needed columns.

- Using joins instead of subqueries.

- Filtering data early using WHERE clauses.

- Avoiding functions on indexed columns in WHERE clauses.

- Using query execution plans to identify bottlenecks.

28. Explain the difference between a clustered and a non-clustered index.

- A clustered index determines the physical order of data in a table and only one can exist per table. A non-clustered index creates a separate structure for the index and can be multiple per table.

29. How do you perform a database backup?

- Backup methods vary by RDBMS but generally involve:

- Using database-specific commands (e.g., `mysqldump` for MySQL, `pg\_dump` for PostgreSQL, `BACKUP DATABASE` in SQL Server).

- Scheduling regular automated backups.
- Ensuring both full and incremental backups as needed.

30. What is ACID compliance?

- ACID stands for Atomicity, Consistency, Isolation, Durability:
- Atomicity ensures transactions are all-or-nothing.
- Consistency ensures transactions bring the database from one valid state to another.
- Isolation ensures concurrent transactions do not interfere with each other.
- Durability ensures that once a transaction is committed, it will remain so.

31. How do you handle database deadlocks?

- Deadlocks can be handled by:
- Implementing deadlock detection algorithms.
- Setting transaction timeouts.
- Using retry logic in applications.
- Ensuring transactions acquire locks in a consistent order.

32. What is a transaction log and how is it used?

- A transaction log records all transactions and changes to the database.

It's used for:

- Recovery of the database in case of failure.
- Rollback of incomplete transactions.
- Replication and auditing purposes.

33. Explain the concept of database sharding.

- Sharding is a database partitioning technique where data is horizontally partitioned across multiple database servers. Each shard contains a subset of the data, allowing for distributed storage and processing.

34. How do you secure a database?

- Database security can be ensured by:
 - Implementing encryption for data at rest and in transit.
 - Using strong authentication and authorization mechanisms.
 - Regularly applying patches and updates.
 - Monitoring and auditing database activity.
 - Implementing network security measures.

35. What is the difference between replication and mirroring?

- Replication involves copying and maintaining database objects in multiple databases, allowing for data distribution and redundancy. Mirroring is maintaining an exact copy of a database on a different server, usually for high availability and disaster recovery.

36. Describe the process of database normalization and its importance.

- Database normalization involves structuring a database to reduce redundancy and improve data integrity. It ensures that each piece of data is stored only once, reducing the likelihood of data anomalies and improving efficiency.

37. What is a materialized view?

- A materialized view is a database object that contains the results of a query. Unlike a regular view, the data is stored on disk and can be refreshed periodically. It improves query performance by pre-computing expensive operations.

38. How do you monitor database performance?

- Database performance can be monitored using:
 - Performance counters and metrics.
 - Query execution plans.
 - Database-specific tools (e.g., SQL Server Profiler, MySQL Performance Schema).
 - Alerts and monitoring software.

39. What are stored procedures and how are they used?

- Stored procedures are precompiled SQL code that can be executed on the database server. They improve performance by reducing the amount of data transfer between applications and the database and enhance security by encapsulating logic.

40. How do you implement database partitioning?

- Database partitioning can be implemented by:
 - Range partitioning: dividing data based on a range of values.
 - List partitioning: dividing data based on a list of values.
 - Hash partitioning: dividing data using a hash function.
 - Composite partitioning: combining multiple partitioning methods.

41. Explain the concept of an execution plan.

- An execution plan is a detailed description of how a database will execute a query, including the order of operations, index usage, and join methods. It helps in optimizing queries by identifying bottlenecks and inefficient operations.

42. What is the role of a DBA in database recovery?

- A DBA is responsible for:
 - Performing regular backups.
 - Implementing and testing recovery plans.
 - Restoring databases in case of failure.
 - Ensuring data integrity during recovery processes.

43. How do you handle large databases in terms of maintenance and performance?

- Handling large databases involves:
 - Regular maintenance tasks such as indexing, statistics updates, and cleanup.
 - Using partitioning and sharding.
 - Archiving old data.
 - Implementing efficient query designs and optimization techniques.

44. What is SQL injection and how can it be prevented?

- SQL injection is a security vulnerability that allows an attacker to execute arbitrary SQL code. It can be prevented by:
 - Using parameterized queries and prepared statements.
 - Validating and sanitizing user inputs.
 - Using stored procedures.

45. Explain the concept of database clustering.

- Database clustering involves using multiple servers to host a single database, providing high availability and load balancing. It ensures that if one server fails, others can take over without downtime.

46. How do you handle data migration between different database systems?

- Data migration involves:
 - Planning and mapping source and target schemas.
 - Extracting, transforming, and loading (ETL) data.

- Testing the migration process.
- Ensuring data integrity and consistency.

47. What are the differences between OLTP and OLAP databases?

- OLTP (Online Transaction Processing) databases are designed for transactional workloads, with fast query processing and maintaining data integrity. OLAP (Online Analytical Processing) databases are optimized for querying and reporting, handling large volumes of data and complex queries.

48. How do you perform database tuning?

- Database tuning involves:
 - Optimizing queries and indexes.
 - Configuring hardware and software settings.
 - Monitoring and adjusting database parameters.
 - Regular maintenance tasks like defragmentation and updating statistics.

49. Explain the concept of a database schema.

- A database schema is the structure that defines the organization of data within a database, including tables, views, indexes, and relationships. It serves as a blueprint for how data is stored and accessed.

50. What are the common causes of database corruption and how do you prevent them?

- Common causes of database corruption include hardware failures, software bugs, and human errors. Prevention methods include:

- Regular backups and testing recovery procedures.
- Using reliable hardware and performing regular maintenance.
- Applying patches and updates to database software.
- Monitoring and auditing database activities.

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